

LECTURE-30 DISCRETE TIME FOURIER SERIES



Discrete Time Fourier Series

- It is a tool for analysis of infinite duration discrete time periodic signals.
- computed for discrete time periodic signals
- DTFS is the only Fourier representation having finite number of discrete components in both time & freq. domain. **SN

DT Fourier Series is given by

$$x[n] = \sum_{k=0}^{N-1} C_k e^{jk\omega_0 n}$$

N: period of DT signal ; $\omega_0 = 2\pi/N$
 ω_0 : fund. freq.

Fourier Series coefficients are given by,

$$C_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\omega_0 n}$$

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Difference b/w CTFS & DTFS

- DTFS is a finite series (has finite no. of coeff) while CTFS is an infinite series.
- Since, in discrete domain there are finite numbers in a sequence
∴ there are no convergence issues
no Dirichlet condⁿ required
- DTFS coefficients are periodic while CTFS coefficients are aperiodic.

Property of DTFS

1. Linearity

let $x[n]$ & $y[n]$ be 2 signals having same period N_0

$$x[n] \longrightarrow c_k$$

$$y[n] \longrightarrow d_k$$

$$\text{then } z[n] = ax[n] + by[n] \longrightarrow ac_k + bd_k$$

2. Time Shifting

$$x[n] \longrightarrow C_k$$

$$x[n-n_0] \longrightarrow e^{-jk\omega_0 n_0} C_k$$

3. Frequency Shifting

or multiplication with an exponential

$$x[n] \longrightarrow C_k$$

$$e^{j\omega_0 n} x[n] \longrightarrow C_{k-m}$$

Time Domain

Freq Domain

Shift

exponential multiply

exponential multiply

shift

4. Time Reversal

$$x[n] \longrightarrow C_k$$

$$x[-n] \longrightarrow C_{-k}$$

- reversal in time domain leads to reversal in freq. domain or reversal of FS coefficients.

$$x[n]: \text{ even} \quad x[-n] = x[n] \quad C_{-k} = C_k : \text{ even}$$

$$x[n]: \text{ odd} \quad x[-n] = -x[n] \quad C_{-k} = -C_k : \text{ odd}$$

nature of DTFS coefficient is same as
nature of $x[n]$

5. Time scaling

$$x[n] \longrightarrow C_k$$

$$x\left[\frac{n}{\alpha}\right] \longrightarrow C_{\alpha k} : \text{fundamental freq. remain same}$$

• we cannot relate coefficient of decimated signal to original signal
Hence, time scaling is allowed only for interpolation.

$$C_k = \sum_{n=0}^{N-1} x[n] e^{-jk\omega_0 n}$$

$$d_k = \sum x\left[\frac{n}{\alpha}\right] e^{-jk\omega_0 n} \Rightarrow n/\alpha = m$$

$$x\left[\frac{n}{2}\right] \longrightarrow C_{2k}$$

$$K=0 \quad C_0$$

$$K=1 \quad C_2$$

$$K=2 \quad C_4$$

6. Multiplication in Time Domain

$$x[n] \longrightarrow c_k$$

$$y[n] \longrightarrow d_k$$

$$\text{then } x[n] \cdot y[n] \longrightarrow c_k * d_k$$

periodic convolution

$$\sum_{m=0}^{N-1} c_m d_{k-m}$$

But in CTFs coeff are aperiodic

$$\sum_{m=-\infty}^{\infty} c_m d_{k-m}$$

7. Convolution in Time Domain

$$x[n] \longrightarrow c_k$$

$$y[n] \longrightarrow d_k$$

$$x[n] * y[n] = \sum_{\langle N \rangle} x[k] y[n-k] \longrightarrow N c_k d_k$$

8. Conjugate

$$x[n] \longrightarrow C_k$$

$$x^*[n] \longrightarrow C_{-k}^*$$

$$x[n] : \text{real} \quad x[n] = x^*[n]$$

$$C_k = C_{-k}^* : \text{conjugate symmetric}$$

$$x[n] : \text{imag} \quad x^*[n] = -x[n]$$

$$C_{-k}^* = -C_k$$

$$\text{conjugate anti symmetric}$$

Same tables expressing relation b/w nature of time domain signal & FS coeff are applicable.

9. Parseval's Relationship

$$x[n] \longrightarrow c_k$$

power of signal: $x[n]$

$$P_x = \frac{1}{N} \sum_{\langle N \rangle} |x[n]|^2$$

$$= \sum_{\langle N \rangle} |c_k|^2$$

↳ consider only one period of coefficients

10. 1st difference property

$$x[n] \longrightarrow C_k$$

$$y[n] = x[n] - x[n-1] \longrightarrow C_k - e^{-jk\omega_0} C_k = C_k (1 - e^{-jk\omega_0})$$

11. Periodicity

$$x[n] \longrightarrow C_k$$

$x[n]$ is periodic with period N

$$x[n+N] = x[n]$$

C_k is also periodic with same period

$$C_{k+N} = C_k$$

Question-01

Complete Fourier series coefficient of $x[n] = \sin [0.2\pi n]$

$$2\pi/\omega_0 = 10 \quad N=10$$

$$\omega_0 = 0.2\pi$$

$$x[n] = \sin(0.2\pi n) = \frac{e^{j0.2\pi n} - e^{-j0.2\pi n}}{2j}$$

$$= \frac{1}{2j} e^{j\omega_0 n} - \frac{1}{2j} e^{-j\omega_0 n} = \sum c_k e^{-jk\omega_0 n}$$

↓

$$k=1$$

$$c_1 = 1/2j$$

↓

$$k=-1$$

$$c_{-1} = 1/2j = c_{-1+10} = c_9$$

Question-02

Complete power of given signal by Parseval's relationship.

$$x[n] = \cos\left(\frac{n\pi}{5}\right) + \sin\left(\frac{n\pi}{6}\right)$$

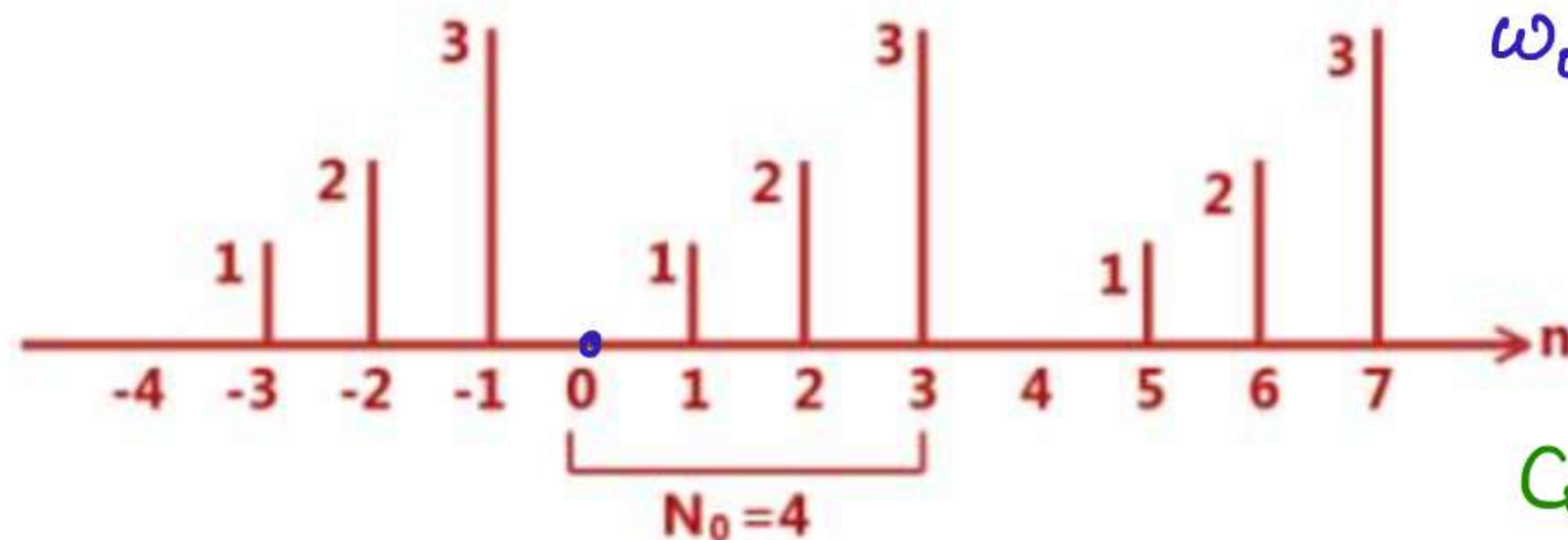
$N_1: 2\pi/\pi/5 = 10$ $N_2 = 2\pi/\pi/6 = 12$
 $N = \text{LCM}(10, 12) = 60$, $\omega_0 = 2\pi/60 = \pi/30$

$$\begin{aligned} &= \frac{1}{2} e^{jn\pi/5} + \frac{1}{2} e^{-jn\pi/5} + \frac{1}{2j} e^{jn\pi/6} - \frac{1}{2j} e^{-jn\pi/6} \\ &= \frac{1}{2} e^{jn6\pi/30} + \frac{1}{2} e^{jn(-6)\pi/30} + \frac{1}{2j} e^{jn(5)\pi/30} - \frac{1}{2j} e^{jn(-5)\pi/30} \\ &\quad \downarrow \qquad \qquad \downarrow \qquad \qquad \downarrow \qquad \qquad \downarrow \\ &\quad c_6 \qquad \qquad c_{-6} \qquad \qquad c_5 \qquad \qquad c_{-5} \end{aligned}$$

$$\begin{aligned} P = \sum |c_n|^2 &= \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 \\ &= 1 \end{aligned}$$

Question-03

Calculate Fourier series coefficient of given signal $x[n]$



$$N_0 = 4$$

$$\omega_0 = \frac{2\pi}{N_0} = \pi/2$$

$$C_k = \frac{1}{4} \left(e^{-j k \pi/2} + 2e^{-j k \pi} + 3e^{-j \frac{3\pi k}{2}} \right)$$

$$C_0 = \frac{e^{-0} + 2e^{-0} + 3e^{-0}}{4} = 1.5$$

$$\begin{aligned} C_1 &= \frac{1}{4} \left(e^{-j \pi/2} + 2e^{-j \pi} + 3e^{-j 3\pi/2} \right) \\ &= \frac{1}{4} (-j - 2 + 3j) \\ &= \frac{-1 + j}{2} \end{aligned}$$

$$C_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-j k \omega_0 n}$$

$$= \frac{1}{N} \sum_{n=0}^{3} x[n] e^{-j k \pi/2 n}$$

$$= \frac{1}{4} \left[x[0] e^{-j k \pi/2 (0)} + x[1] e^{-j k \pi/2 (1)} \right. \\ \left. + x[2] e^{-j k \pi/2 (2)} + x[3] e^{-j k \pi/2 (3)} \right]$$

$x[n]$: real

$$C_{-1}^* = C_1 \leftarrow \text{conj symmetry}$$

$$C_{-1} = C_1^* = -\frac{1-j}{2}$$

$$C_3 = C_{-1} \quad (N=4) \quad C_{-1+4} = C_3$$

$$C_3 = -\frac{1-j}{2}$$

$$C_2 = \frac{1}{4} [e^{-j\pi} + 2e^{-j2\pi} + 3e^{-j\pi}]$$

$$= \frac{1}{4} (-1 + 2 - 3) = -\frac{1}{2}$$

$$\left\{ \dots, \frac{3}{2}, -\frac{1+j}{2}, -\frac{1}{2}, -\frac{1-j}{2}, \frac{3}{2}, \dots \right\}$$

↑

Question-04

Calculate Fourier series coefficient C_k of given signal $x[n]$



$$N_0 = 10 \quad \omega_0 = 2\pi/N_0 = \pi/5$$

$$C_k = \frac{1}{10} \left(\frac{1 - e^{-j5k \times \pi/5}}{1 - e^{-jk\pi/5}} \right)$$

$$C_k = \frac{1}{10} \sum_{n=0}^9 x[n] e^{-jk\omega_0 n}$$

$$= \frac{1}{10} \sum_{n=0}^4 1 \cdot e^{-jk\omega_0 n}$$

$$= \frac{1}{10} \times \frac{1 - (e^{-jk\omega_0})^5}{(1 - e^{-jk\omega_0})}$$

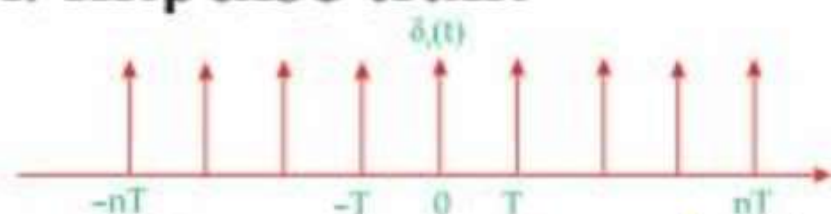
$$C_k = \frac{1}{10} \left(\frac{1 - e^{-jk\pi}}{1 - e^{-jk\pi/5}} \right)$$

Question-05

Match list-I (Name of periodic function) with the list-II (Properties of spectrum function) and select the correct answer using the codes given below the lists:

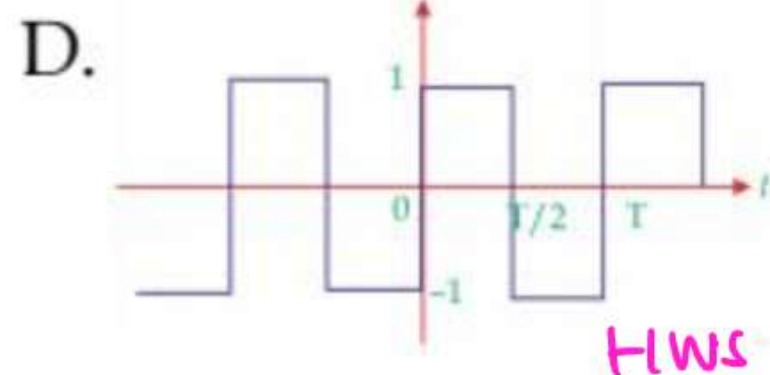
list-I (Name of periodic function)

A. Impulse train



B. Full wave rectified sine

C. $2 \sin \frac{2\pi t}{6} \cos \frac{4\pi t}{6} = \sin \pi t - \sin \frac{2\pi t}{3}$
 $\omega_1 = \pi$ $\omega_2 = \pi/3$
 $\omega_0 = \pi/3$



list-II (Properties of spectrum function)

1. Only even harmonics are present

2. Impulse train with strength $1/T$

3. $C_3 = \frac{1}{2j}; C_{-3} = -\frac{1}{2j}$

4. Only odd harmonics are present

Code: A

B

C

D

(a) 5

2

3

1

(b) 2

5

3

4

(c) 5

2

4

3

~~(d) 2~~

1

3

4

Question-06

The Fourier series coefficient C_n of $x(t)$ are

~~(a)~~ $\frac{1}{2}(\delta[n+1] + \delta[n-1])$

(b) $\frac{1}{2}(\delta[n+1] - \delta[n-1])$

(c) $\frac{1}{2}(\delta[n-1] - \delta[n+1])$

(d) None of these

$$x(t) = \cos t$$

$$y(t) = \sin t$$

$$z(t) = x(t) \cdot y(t)$$

$$\begin{aligned} x(t) = \cos t &= \frac{1}{2} e^{jt} + \frac{1}{2} e^{-jt} \\ &\quad \downarrow \qquad \qquad \downarrow \\ &\quad c_1 \qquad \qquad c_{-1} \\ &\quad n=-1 \qquad n=1 \\ &\quad \left\{ \frac{1}{2}, 0, \frac{1}{2} \right\} \end{aligned}$$

Question-07

The Fourier series coefficient of $y(t)$, d_n will be

(a) $\frac{1}{2}(\delta[n+1] + \delta[n-1])$

~~(b) $\frac{j}{2}(\delta[n+1] - \delta[n-1])$~~

(c) $\frac{j}{2}(\delta[n-1] - \delta[n+1])$

(d) $\frac{1}{2j}(\delta[n+1] + \delta[n-1])$

$$\frac{1}{j} = \frac{j}{j^2} = \frac{j}{-1} = -j$$

$$y(t) = \sin t = \frac{1}{2j} e^{jt} - \frac{1}{2j} e^{-jt}$$

$$C_1 = \frac{1}{2j} \quad C_{-1} = -\frac{1}{2j}$$

$$\left\{ \frac{-1}{2j}, \underset{\uparrow}{0}, \frac{1}{2j} \right\} = \left\{ \underset{n=-1}{\frac{1}{2}}, \underset{\uparrow}{0}, \underset{n=1}{-\frac{j}{2}} \right\}$$

Question-08

The Fourier series coefficient of $z(t)$, e_n will be

- (a) ~~$\frac{1}{4j}(\delta[n-2] - \delta[n+2])$~~ (b) $\frac{1}{2j}(\delta[n-2] - \delta[n+2])$
(c) $\frac{1}{2j}(\delta[n+2] - \delta[n-2])$ (d) None of these

$$x(t) \cdot y(t) \rightarrow C_k * d_k$$

$$z(t) = x(t) \cdot y(t)$$

$$= \cos t \sin t$$

$$= \frac{1}{2} \sin 2t$$

$$= \frac{1}{4j} e^{j2t} - \frac{1}{4j} e^{-j2t}$$

$$C_2 = \frac{1}{j4}$$

$$C_{-2} = -\frac{1}{j4}$$

Question-09

$$\omega_0 = 2\pi/T = \pi/4$$

$$C_{-1} = (j)^* = -j$$

$$C_{-5} = 2^* = 2$$

A continuous time periodic signal has a fundamental time period $T = 8$. The non-zero Fourier series coefficients are as, $c_1 = c_{-1}^* = j$, $c_5 = c_{-5}^* = 2$. The signal will be

~~(a)~~ $4 \cos\left(\frac{5\pi}{4}t\right) - 2 \sin\left(\frac{\pi}{4}t\right)$

(b) $2 \cos\left(\frac{\pi}{4}t\right) + 4 \sin\left(\frac{5\pi}{4}t\right)$

(c) $2 \cos\left(\frac{5\pi}{4}t\right) + 2 \sin\left(\frac{\pi}{4}t\right)$

(d) $-2 \cos\left(\frac{\pi}{4}t\right) + 4 \sin\left(\frac{5\pi}{4}t\right)$

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega_0 t}$$

$$= C_{-5} e^{-j5\omega_0 t} + C_{-1} e^{-j\omega_0 t} + C_1 e^{j\omega_0 t} + C_5 e^{j5\omega_0 t}$$

$$= 2 e^{-j5\pi/4 t} - j e^{-j\pi/4 t} + j e^{j\pi/4 t} + 2 e^{j5\pi/4 t}$$

$$= 4 \cos\left(\frac{5\pi}{4}t\right) + j(2j) \sin\left(\frac{\pi}{4}t\right)$$

Question-10

The FS coefficient of time-domain signal $x(t)$ is $c_n = j \delta[n-1] - j \delta[n+1] + \delta[n+3] + \delta[n-3]$. The fundamental frequency of signal is $\omega_0 = 2\pi$. The signal is

(a) $2(\cos 3\pi t - \sin \pi t)$

(b) $-2(\cos 3\pi t - \sin \pi t)$

~~(c) $2(\cos 6\pi t - \sin 2\pi t)$~~

(d) $-2(\cos 6\pi t - \sin 2\pi t)$

$$C_n = \{1, 0, -j, 0, j, 0, 1\}$$

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$= C_{-3} e^{-j3\omega_0 t} + C_{-1} e^{-j\omega_0 t} + C_1 e^{j\omega_0 t} + C_3 e^{j3\omega_0 t}$$

$$= e^{-j6\pi t} - j e^{-j2\pi t} + j e^{j2\pi t} + e^{j6\pi t}$$

$$= 2\cos 6\pi t + j(2j \sin 2\pi t)$$